RealView® System Generator

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Agenda

- The software defined product
  - What is all this software?
  - What is the problem?
  - What is the solution?
  - What is needed to be successful?
- The product
  - What does it look like?
  - How does it work?
- Customer success
- How can I win more business?
- Summary

The Software Defined Product

- New class of emerging products beginning to ship
  - Home media servers
  - Portable media players (video, image, music)
  - MPEG-based camcorders
- These products demand
  - Higher performance
  - More interaction
  - Mobile commerce

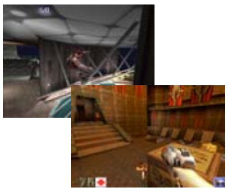
- Technologies
  - Video compression
  - 2D/3D graphics
  - Security
  - Power management
  - Biometrics
  - Cryptography
  - Sensor processing
  - Audio compression

Consumer Electronic Product Success

- Vendors need
  - Feature-rich differentiated products
- Which require
  - Testing and verification of hardware and software integration
  - Content generation
  - Validation of the end-user experience




Example – 3D Games



- **Turnaround time** for test is lengthy due to differing interface standards and slow programming on real devices
- All games must go through **QA at the distributor**, which is expensive
- Problems are very difficult to **diagnose** on the real device
- Very costly for the operator if not available **in time**

- **Games must run on multiple platforms to be cost effective**

Example – Secure Services

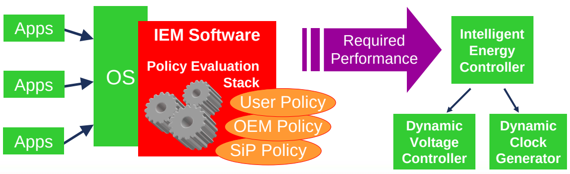
- Mobile commerce is on the rise
  - Need for secure transactions
  - Prevention of identity theft
- Mobile devices used as business platforms
  - Need for data privacy
  - Prevention of data corruption
  - Guarding against viruses
- Home entertainment
  - Operators need to protect revenue streams
  - Content needs to be shared around multiple devices in the home
- **Need to test how people interact with these services**

Example – Mobile TV

- How will the TV picture scale for the end device?
  - Can you read text?
  - Can you see the faces of presenters?
- How quickly can you browse the channels?
- How does TV interact with other functions, such as an incoming call?
- User interface and speed are paramount
- **Test environment needs real-time response**

Example – Intelligent Energy Manager

- IEM technology is a hardware and software solution for energy management
  - Dynamic control of voltage and frequency
- IEM software collects task information
  - Prediction of future performance requirement is made
- Appropriate operating point (frequency and voltage) is determined
- Applications can signal energy requirements through user policies
- **Need to test multiple scenarios on latest ARM cores to tune user policies**



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What is the Problem?

- Real devices and emulation hardware not available early enough in the product lifecycle

- This means
  - Lack of content at product launch
  - Integration problems found late in the day
  - Poor user experience
- **However, content developers don't always need the accuracy of real hardware because they write code to standard interfaces (APIs)**

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What is the Solution?

- A virtual platform
  - Easy to deploy via a Web portal
  - Easy to upgrade and maintain
  - No costs associated with shipping hardware
  - No special debug hardware required to connect to development workstation
  - Easy to modify and expand by software developers
  - Easy to understand by non-hardware engineers
- Essentials
  - The speed is representative of the real device
  - Tools must be provided to build, configure and debug the platform

What is Needed to be Successful?

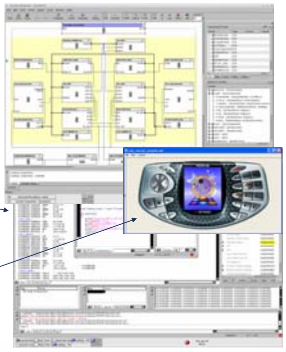
- Two basic requirements
  - An environment for creating virtual platforms around processor models
  - Very fast processor models to interact with in real time

RealView®

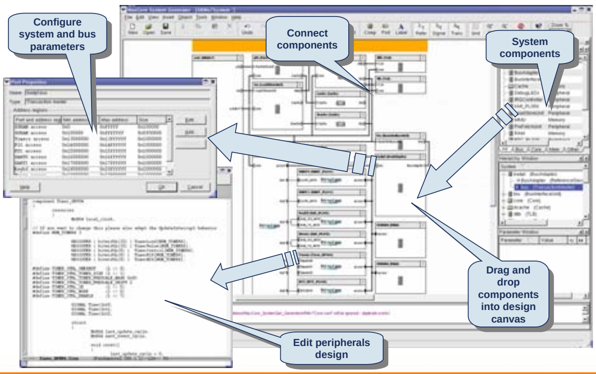
System Generator and Code Translation Models

The Development Environment

- A block diagram-based tool that can select building blocks from a library
  - RealView System Generator
- An industry-leading compiler to create efficient code
- A software debugger to find code errors and tune performance
  - RealView Development Suite
- An interactive graphical representation of the final device
  - RealView Real Time System Model



Look of the System Generator Tool



System Generator Tool Demonstration

- Graphical design entry
  - Drag and drop with components
  - Establishing connections
  - Adding and editing custom IP
  - Checking design
- Generating model
- Connecting debugger
- Running application code

Certified High-Speed ARM IP Models

- Only ARM owns the verification IP to certify correctness
  - 1000s of test programs developed for hardware validation
- Models include
  - TrustZone® security technology
    - Run secure services on a real device to check user interaction
  - Jazelle® Java acceleration technology
    - Run Java applets on a virtual machine on top of a platform OS, all in real time
  - Intelligent Energy Manager technology
    - Run many applications in an OS context to help tune user policies

What Makes Them So Fast?

- Code translation technology
  - Translates ARM assembly code instructions dynamically into instructions that run on the development workstation
- System peripherals are only activated when necessary
- Host computer provides
  - Display
  - Keyboard input
  - Network connections
  - Sound output
  - Camera
  - Biometrics

```
   MOV r0, #20
L1 SUBS r0, r0, #1
   BNE L1
```

ARM Code

```
   mov    eax, 20
   mov    dword ptr [state+REG_R0], eax
L1 mov    eax, dword ptr [state+REG_R0]
   sub    eax, 1
   setnb  byte ptr [state+FLAGS_C]
   seto   byte ptr [state+FLAGS_V]
   setz   byte ptr [state+FLAGS_Z]
   sets   byte ptr [state+FLAGS_N]
   mov    dword ptr [state+REG_R0], eax
   testb  byte ptr [state+FLAGS_Z], 1
   je     $1
   b      L1
$1
```

PC Code

How Does It Work?

Original ARM code

```
   MOV r0, #20
L1 SUBS r0, r0, #1
   BNE L1
```

Translated to function calls

```
   imm = dataproc_imm_op2();
   set_reg(rd, imm);
L1 v0 = dataproc_imm_op1(rn);
   imm = dataproc_imm_op2();
   sub(v0, imm);
   set_flags_sub();
   set_reg(rd, v0);
   compile_cond(NE, $1);
   branch_back(L1);
$1
```

Generated IA-32 (X86) code

```
   mov    eax, 20
   mov    dword ptr [state+REG_R0], eax
L1 mov    eax, dword ptr [state+REG_R0]
   sub    eax, 1
   setnb  byte ptr [state+FLAGS_C]
   seto   byte ptr [state+FLAGS_V]
   setz   byte ptr [state+FLAGS_Z]
   sets   byte ptr [state+FLAGS_N]
   mov    dword ptr [state+REG_R0], eax
   testb  byte ptr [state+FLAGS_Z], 1
   je     $1
   b      L1
$1
```

Optimized X86 code

```
   mov    eax, 20
L1 sub    eax, 1
   jne    L1
   setnb  byte ptr [state+FLAGS_C]
   seto   byte ptr [state+FLAGS_V]
   setz   byte ptr [state+FLAGS_Z]
   sets   byte ptr [state+FLAGS_N]
   mov    byte ptr [state+REG_R0], eax
```

How Does It Work?

- Original code executes in 41 instructions
  - Turned by the decoder into a sequence of function calls which emit code
- This is 201 instructions executed, including 101 stores and 21 loads
  - The processor optimizes some of this away
- With data flow analysis, this could be further optimized
- Final code is 46 instructions with at most 5 stores

Preliminary Benchmark Results

- Benchmark system: Pentium-M, 1.7GHz, 1GB RAM, running Linux under VMware

Operation	Demonstrates	Measured
Dual-core Linux boot: u-boot + 2.6.14 kernel + 9MB ROMFS	Realistic scenario	7s or ~128M i/s (~64M per core)
Both cores idle at Linux prompt	Peak performance	~570M i/s (285M per core)
Both cores executing 'ls –ltR'	Performance with repeated I/O	~152M i/s (76M per core)
Both cores executing gzip / gzip –d on a file in RAMFS	Compute-bound performance	~124M i/s (62M per core)
One core idle, other core gzip / gzip –d on file in RAMFS	Asymmetric performance	~200M i/s (100M per core)

Customer Success

- IdeaWorks3D
  - Save 20% of test and development time
  - Less reliance on real devices, which are a scarce commodity
- Compiler team
  - Save hours on regression tests
- TrustZone software deployment
  - Reduced costs of demonstration system

How Can I Win More Business?

- Demonstrate product concepts at an early stage
  - A real simulation is much more compelling than PowerPoint or a mock-up
- No need to model the hardware accurately
  - Abstract simulation enables near real-time simulation speed
- User interaction is all important
- Styling as important as the technology

Summary

- RealView System Generator virtual prototypes interact with the environment in real time
  - Much faster than traditional software models
- Software engineers can generate virtual prototypes themselves
  - No need to pay for consulting services
- Content generators can use the virtual prototypes to run and test their creative output
  - Until now, they could only “emulate” a device on a PC, not simulate the real processor
- The only certified high-speed models of ARM IP
  - ARM’s ability to certify models is a key differentiator

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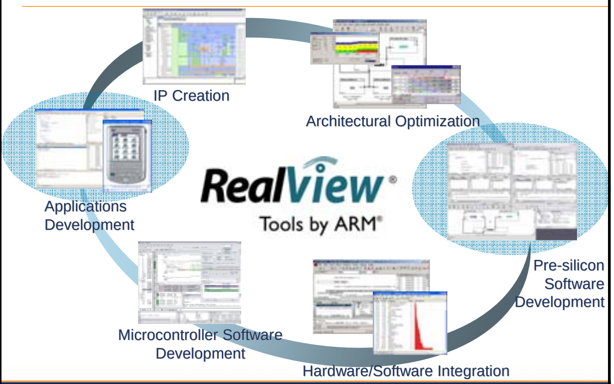
RealView Tools for the Complete Solution



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